WORKING WITH *Viking!*

HOW *Viking!* WORKS

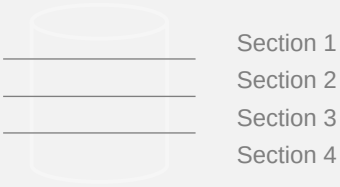
Viking! is a program designed to reconstruct a 3-dimensional image of the connectome.

1. The tissue is isolated and processed for electron microscopy (EM).
2. The processed tissue is then cut into very thin slices.
3. The slices are digitally imaged using transmission electron microscopy (TEM). To acquire high-resolution images, small areas of the field are imaged at high magnification and a montage of the entire field is created.
4. The slices are then digitally fitted together to make a complete volume.
5. *Viking!* allows cells to be reconstructed throughout the depth of the volume in order to create accurate 3D images.

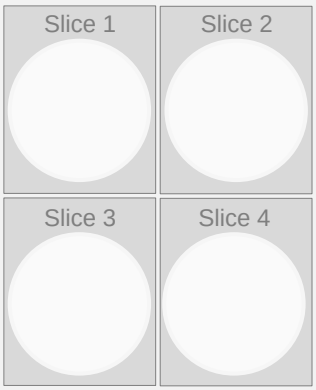
THE BASICS

What cell looks like intact

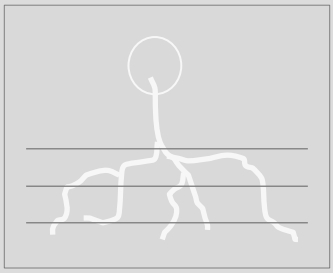
If the cell is sectioned at regular intervals!



Slices will look like this when viewed from the top



Viking! allows the user to see all of the cells in a given volume of tissue. Keep in mind that orientation in 3-dimensional space is highly variable, so the portions of cells visible as images in *Viking!* will have many different shapes.



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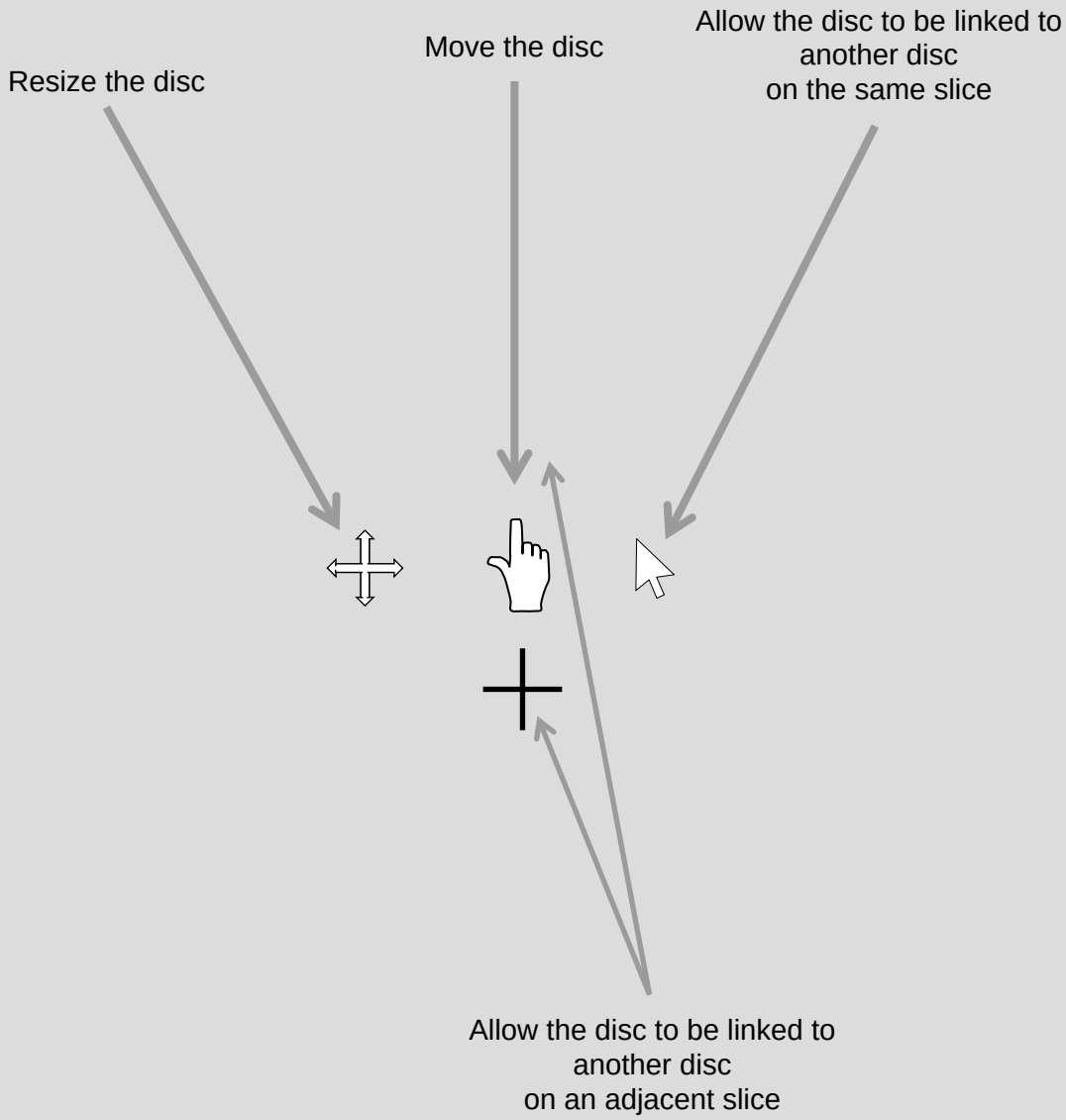
CURSOR POSITIONS WITHIN DISCS

Placing the cursor over different locations within a disc will (cursor icons enlarged for clarity):



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SUMMARY:

- Cells are **parents.**
- Synapses, gap junctions, etc. are **children** belonging to their associated parent.
- To annotate a child structure, the associated parent must be annotated.
- Each child structure must be linked to a parent.
- Annotate child structures in all slices in which they appear.

HOW TO LINK CHILD ANNOTATIONS (ILLUSTRATED)

1. If you are annotating a child structure, first add the appropriate structure disc. After setting the disc, a colored band will appear. Without clicking, move the cursor to the center of the parentÕs disc and left click once to link the two discs.

2. For synapses, adherens, gap junctions, cisterns, etc., you must specify the direction of the communication.
   - Place the cursor inside the disc of the pre- partner and left click once
   - Drag the cursor to the disc of the post- partner, then left click once
   - An animated white band will appear and move in the specified direction of the interaction.

For bidirectional annotations, place the cursor between the partner discs, right-click twice, and select Bidirectional.

- Directional: conventional, bipolar conventional, ribbon, cistern, etc.
- Bidirectional: gap junction, adherens, etc.

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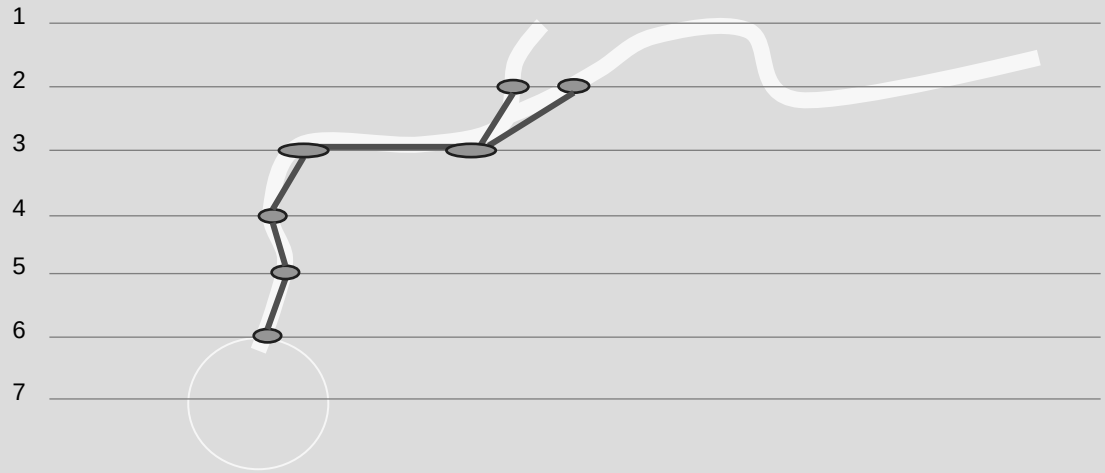
HOW TO ANNOTATE A BRANCHING CONTOUR

You have now generated this:

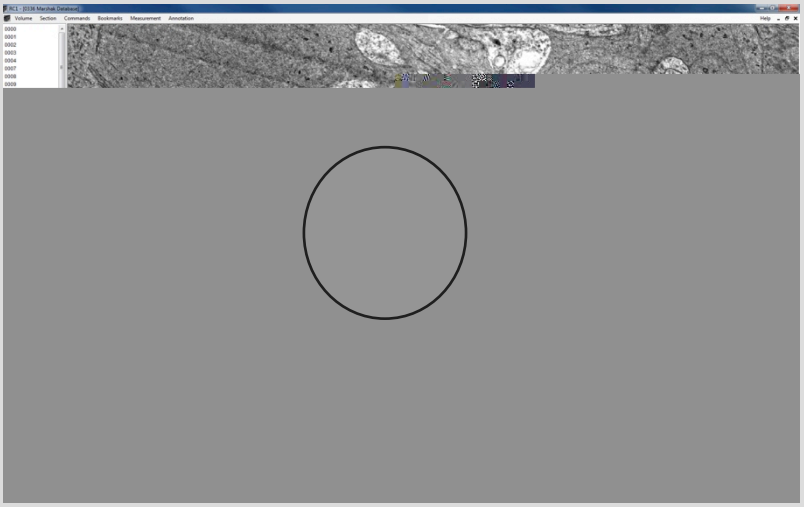


Both branches may now be annotated as usual.

This scenario highlights the need to survey a contour before annotating so that one formulates a plan for annotation and avoids surprises

ANNOTATION DISC

Annotate at the center of mass of a cell without exceeding the bounds of the plasma membrane Ð even for oddly-shaped or uneven cells. For example, the irregular cell seen below would be annotated as follows:



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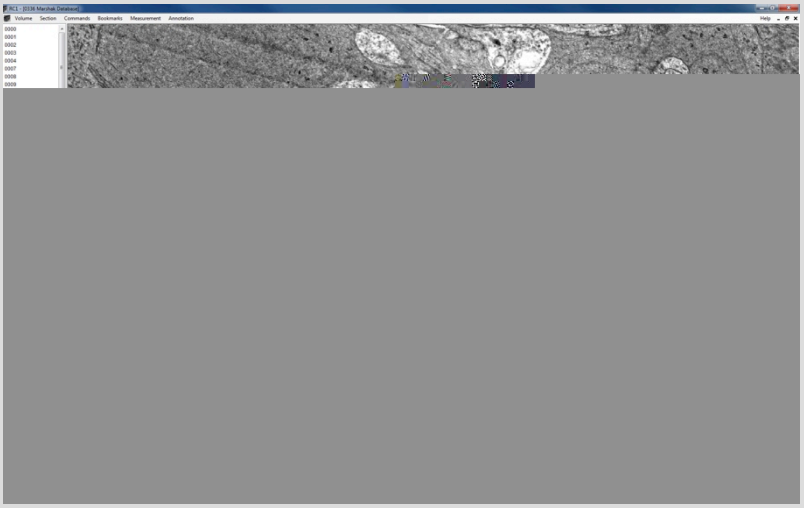
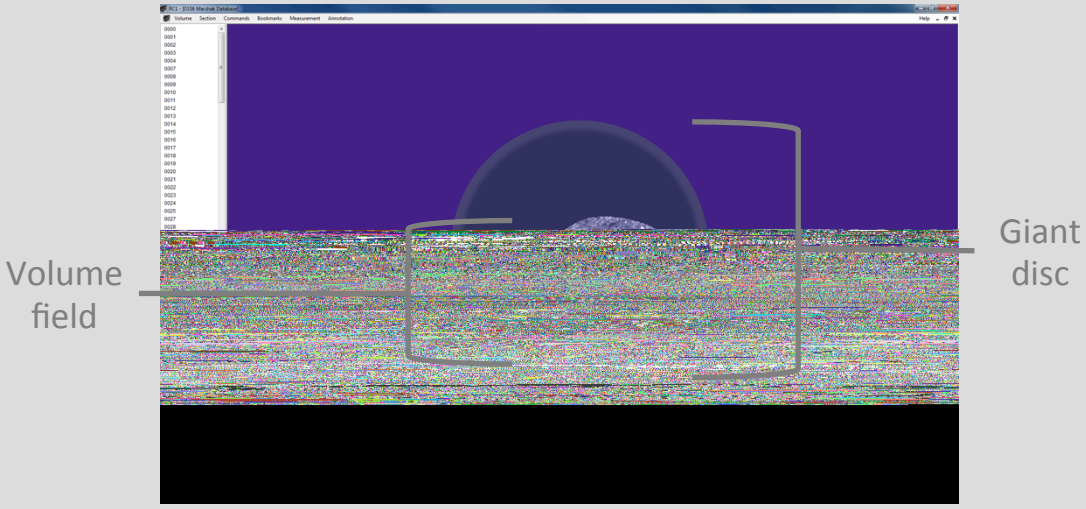
WHAT TO DO IF YOU SEE TWO ARROWHEADS BUT NO DISC OUTLINE

Sometimes you may see one or two arrowheads indicating the presence of a disc, but be unable to see the outline of the disc. In addition, the field may have a grey cast to it.



This is an indication that the disc is *very* large.

Simply zoom out and resize the disc (or delete if it was placed by mistake).